Marked coordinates and exact orbit enumeration

human_compression research project

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1 Coordinates on a marked group

A marking is an ordered tuple, not an unordered generating set. The distinction matters whenever the positions specify face ranks. Let G be a group and let $\boldsymbol{s} = (s_1, \dots, s_m)$ be a tuple in G. Define

$$a_0 = 1, \qquad a_i = s_1 \cdots s_i \quad (1 \le i \le m). \tag{1}$$

The reverse substitution is $s_i = a_{i-1}^{-1} a_i$. For a set $J \subseteq \{0, \dots, m\}$ put

$$D_J(\boldsymbol{a}) = \langle a_i^{-1} a_j : i, j \in J \rangle, \quad T_J(\boldsymbol{s}) = \langle s_{i+1} \cdots s_j : i < j,\ i, j \in J \rangle. \tag{2}$$

An empty generating set generates the trivial subgroup.

Theorem MC.1 (coordinate transport). The substitutions in (1) are mutually inverse, preserve the generated group, commute with every group homomorphism, and satisfy

$$a_i^{-1} a_j = s_{i+1} \cdots s_j \quad (i < j), \qquad D_J(\boldsymbol{a}) = T_J(\boldsymbol{s}). \tag{3}$$

Consequently every assertion involving these particular subgroup diagrams, their intersections, their generated joins and their element orders can be transferred by substitution, with no new group-theoretic hypothesis.

Proof. Adjacent factors cancel in $(a_i^{-1} a_{i+1}) \cdots (a_{j-1}^{-1} a_j)$, leaving $a_i^{-1} a_j$. Conversely, cancellation in $(s_1 \cdots s_i)^{-1} (s_1 \cdots s_j)$ gives the first equality in (3). Each original generator is a word in the new ones and conversely, so generation is preserved. A homomorphism commutes with products and inverses. In (2), a difference with reversed endpoints is the inverse of the difference with increasing endpoints, and equal endpoints contribute the identity. Thus the two displayed generating sets generate exactly the same subgroup. All the asserted transports now follow by replacing equal subgroups or equal elements. □

This elementary substitution is the marked-coordinate principle already recorded as M2 in the project proof bank. Its application to chiral polytopes is the substitution of Buekenhout–Leemans–Tranchida, Theorem 4.3; neither that substitution nor its use of C^+-groups is claimed as new here. The formal certificate accompanying this paper verifies (3) for arbitrary interval lengths and the equality of the actual subgroup closures, including reversed endpoint pairs.

Lemma MC.2 (a star generates the difference subgroup). For nonempty J and $b \in J$,

$$D_J(\boldsymbol{a}) = \langle a_b^{-1} a_j : j \in J \setminus \{b\} \rangle. \tag{4}$$

Proof. The right side is contained in the left. Every generator of the left has the expression $a_i^{-1} a_j = (a_b^{-1} a_i)^{-1} (a_b^{-1} a_j)$, proving the reverse inclusion. □

Formula (4) reduces a subgroup computation with quadratically many named differences to one with at most $|J| - 1$ generators. It changes neither the subgroup nor the intersection tests.

2 Exact equivalence of generating markings

We recall the shared orbit interface used in the enumeration topic. If a group B acts on sets Z, E, and $p : Z \to E$ is equivariant, then

$$B\backslash Z \;\simeq\; \coprod_{e\in B\backslash E} B_e\backslash p^{-1}(e), \tag{5}$$

where one base representative is chosen in each orbit. This is U.ExtensionOrbits. The proof and its previous geometric applications are retained from the shared umbrella: first move the base into the selected orbit representative; any remaining equivalence fixes that base. Formula (5) applies to a tuple and its initial subtuple. In particular, after fixing one entry one must use its stabilizer, rather than continuing to quotient the remaining entries independently by the whole group.

For a finite group G, let $\mathcal{M}_m(G)$ be the set of ordered generating m-tuples. The automorphism group acts freely on this set: an automorphism fixing all the generators fixes every word. Thus every orbit has size $|\operatorname{Aut}(G)|$. One need not construct the whole automorphism group to test this equivalence.

Theorem MC.3 (Cayley transport). Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be generating m-tuples of finite groups G, H of equal order. The assignment $a_i \mapsto b_i$ extends to a group isomorphism if and only if the following finite propagation succeeds. Start with $f(1) = 1$, and whenever $f(x)$ is assigned, require

$$f(xa_i) = f(x)b_i \quad (1 \le i \le m). \tag{6}$$

Reject if two requirements disagree. Accept if the resulting assignment is a bijection. The isomorphism, when it exists, is unique.

Proof. Every group isomorphism with the required generator images satisfies (6), so disagreement or failure of bijectivity is a valid rejection. A finite group generated as a group by the a_i is also generated as a monoid: a_i^{-1} is a positive power of a_i. Hence propagation visits all of G. If it succeeds, induction on a positive word w gives $f(xw(\boldsymbol{a})) = f(x)w(\boldsymbol{b})$. Taking $x = 1$ shows that the right side depends only on the element represented by w. Taking a word for an arbitrary y then gives $f(xy) = f(x)f(y)$. Acceptance therefore gives a bijective homomorphism. Uniqueness follows from generation. □

A canonical implementation starts at the identity, visits new elements in breadth-first order using the ordered generators, and records the target numbers of all colored Cayley edges. Two such finite arrays agree exactly when the markings are isomorphic. Indeed equality of arrays supplies the bijection in MC.3; conversely an isomorphism carries the deterministic traversal to the same traversal. This test does not quotient the tuple by a permutation of its positions.

The same method applies to connected graphs with one edge of each color at every vertex: a color-preserving graph map is forced by the image of one vertex. Trying every image, rejecting inconsistencies and testing bijectivity enumerates the full color-preserving automorphism group. This is a direct finite certificate, not a heuristic based on a chosen subgroup of graph automorphisms.

3 Hereditary searches and termination

Suppose an admissibility predicate is defined on normalized tuples $(1, a_1, \dots, a_m)$ and has the following properties:

1. every initial subtuple is admissible in the subgroup it generates;
2. admissibility is preserved by automorphisms of the ambient group;
3. every appended entry lies in a prescribed finite candidate set;

4. every admissible extension enlarges the subgroup generated by the prefix.

Theorem MC.4 (exhaustive prefix search). Start at representatives of the allowed first-entry conjugacy classes. Recursively append every candidate that passes the admissibility tests. Record a tuple when it generates the input group, and stop that branch. Deduplication by MC.3 gives exactly the automorphism orbits of all admissible generating tuples.

Proof. Every retained tuple is admissible by the tests, and is recorded only when it generates the required group. Conversely, conjugate any admissible generating tuple to put its first entry into the chosen conjugacy representative. Every prefix passes the tests by hypothesis 1; induction on its length shows that the recursion reaches the entire tuple. Hypothesis 4 shows that no proper prefix already generates the ambient group, so stopping a branch early loses no output. The search terminates because each subgroup index along a strict finite subgroup chain is at least two. Finally MC.3 removes exactly, and all, the automorphism equivalences among the recorded tuples. □

The conjugacy reduction need not remove every automorphism equivalence at the first entry. Inner automorphisms suffice for completeness, and final automorphism deduplication removes any remaining duplicates. Alternatively, (5) permits successive reduction by the actual stabilizer of each prefix. Such reduction is optional; its implementation must preserve the prefix rather than treating entries independently.

A useful sufficient condition for strict growth is the difference-subgroup intersection property

$$D_J \cap D_K = D_{J\cap K}. \tag{7}$$

For a prefix $(1, a_1, \ldots, a_{k-1})$ and a nonidentity new entry a_k, take $J = \{0, \ldots, k-1\}$ and $K = \{0, k\}$. Then $\langle a_1, \ldots, a_{k-1}\rangle \cap \langle a_k\rangle = 1$. In particular a_k is not in the old subgroup. If $|a_1| \geq 3$ and the final tuple has n coordinates including a_0, then

$$|G| \geq 3\,2^{n-2}. \tag{8}$$

The implementation may use this bound or simply stop whenever generation occurs. Its proof of termination is independent of a numerical cutoff selected by the programmer.

4 Reflection tests

Let $(1, a_1, \ldots, a_{n-1})$ generate G, assume $a_i^2 = 1$ for $i \geq 2$, and write $s_i = a_{i-1}^{-1} a_i$.

Proposition MC.5 (reflection transport). An automorphism f satisfies

$$f(a_i) = a_i^{-1} \quad (1 \leq i < n) \tag{9}$$

if and only if

$$f(s_1) = s_1^{-1}, \qquad f(s_2) = s_1^2 s_2, \qquad f(s_i) = s_i \quad (i \geq 3). \tag{10}$$

Any such automorphism has square equal to the identity. If $|a_1| > 2$, it is not the identity.

Proof. Condition (9) fixes a_i for $i \geq 2$. Therefore $f(s_2) = f(a_1^{-1}a_2) = a_1 a_2 = a_1^2 s_2$, and both factors in $s_i = a_{i-1}^{-1} a_i$ are fixed for $i \geq 3$. This gives (10). Conversely (10) sends the first prefix to its inverse and fixes the second prefix $s_1 s_2$. All subsequent prefixes are then fixed. These are exactly (9), since they are involutions. Finally f^2 fixes every generating entry, and hence is the identity. Its action on a_1 is nontrivial when $|a_1| > 2$. □

This is the content of BLT Lemma 4.4 in the correct automorphism language, expressed through MC.1. It concerns all group automorphisms, not only conjugations by elements of G. MC.3 decides (9) without requiring the automorphism group to be enumerated in advance.

5 Attribution, formal coverage and reuse

The umbrella collects elementary coordinate and enumeration interfaces. MC.1 is the existing marked-coordinate principle; the specific prefix substitution and reflection transport used here belong to BLT. The orbit-over-a-base result (5) is U.ExtensionOrbits, already used in the accepted enumeration module. MC.2, MC.3 and MC.4 supply explicit complete proofs of the algebraic and algorithmic details needed by applications; no priority is asserted for these elementary statements.

`PrefixCoordinates.lean` proves seven declarations: arbitrary-length telescoping, recovery in both coordinate directions, equality of subgroup closures, transport of the full intersection assertion, the reflected second generator, and the square of an inversion automorphism on an arbitrary generating set. The subgroup theorem is not merely equality of two generated groups at the top: it identifies every subgroup indexed by a set of endpoints. The formal file does not prove the geometric reconstruction theorem, finite search completeness, the involution-count theorem for almost simple groups, or the correctness of the Python interpreter and GAP.

The associated application must pay for its geometric recognition theorem and for its specific admissibility conditions. This umbrella is charged in full to any topic using it under the project's requested topic-local measurement convention. Shared acquisition inside one topic is recorded once; no numerical compression ratio is claimed here.